```
/* ===== FILE: app/layout.tsx ===== */
export const dynamic = 'force-dynamic'

import type { Metadata } from "next";
import { Inter } from "next/font/google";
import { AuthProvider } from "@/lib/auth-context";
import "./globals.css";

const inter = Inter({ subsets: ["latin"] });

export const metadata: Metadata = {
  title: "AI 影视全流程工作流系统",
  description: "从元构思到成片，AI 辅助影视工业化生产",
};

export default async function RootLayout({
  children,
}: Readonly<{
  children: React.ReactNode;
}>) {
  return (
    <html lang="zh-CN">
      <body className={inter.className}>
        <AuthProvider>
          {children}
        </AuthProvider>
      </body>
    </html>
  );
}
/* ===== FILE: app/page.tsx ===== */
export const dynamic = 'force-dynamic'

import Link from 'next/link'
import { getCurrentUser } from '@/lib/auth-helpers'
import { ArrowRight, Sparkles } from 'lucide-react'

export default async function Home() {
  const user = await getCurrentUser()
  const isLoggedIn = !!user

  return (
    <main className="flex min-h-screen flex-col items-cent
    er justify-center p-6 md:p-24 bg-gradient-to-b from-st
    one-50 to-white">
      <div className="max-w-2xl text-center">
        <div className="mb-6 inline-flex items-center gap-
    2 rounded-full bg-amber-50 px-4 py-1.5 text-sm text-am
    ber-700 border border-amber-200">
          <Sparkles className="h-4 w-4" />
```

```
        <span>AI 驱动的影视工业化生产工具</span>
      </div>

      <h1 className="text-4xl md:text-5xl font-bold mb-4
 tracking-tight text-stone-900">
        AI 影视全流程工作流系统
      </h1>

      <p className="text-lg text-stone-500 mb-10 leading
-relaxed">
        从元构思到成片，AI 辅助影视工业化生产
      </p>

      <div className="flex flex-col sm:flex-row items-ce
nter justify-center gap-4">
        {isLoggedIn ? (
          <Link
            href="/dashboard"
            className="inline-flex items-center gap-2 ro
unded-lg bg-stone-900 px-6 py-3 text-sm font-medium te
xt-white transition hover:bg-stone-800 hover:shadow-lg
"
          >
            进入工作台
            <ArrowRight className="h-4 w-4" />
          </Link>
        ) : (
          <>
            <Link
              href="/dashboard"
              className="inline-flex items-center gap-2
rounded-lg bg-stone-900 px-6 py-3 text-sm font-medium
text-white transition hover:bg-stone-800 hover:shadow-
lg"
            >
              开始使用
              <ArrowRight className="h-4 w-4" />
            </Link>
            <span className="text-sm text-stone-400">或<
/span>
            <Link
              href="/login"
              className="inline-flex items-center gap-2
rounded-lg border border-stone-300 bg-white px-6 py-3
text-sm font-medium text-stone-700 transition hover:bg
-stone-50"
            >
              登录账号
            </Link>
          </>
```

```
        )}
      </div>
    </div>
  </main>
 )
}
/* ===== FILE: app/(dashboard)/DashboardShell.tsx ===== */
'use client'

import { useEffect, useState } from 'react'
import AppSidebar from '@/components/layout/AppSidebar'
import Header from '@/components/layout/Header'
import { apiClient } from '@/lib/api-client'

interface User {
  id: string
  name?: string | null
  email?: string | null
  image?: string | null
  isAdmin?: boolean
  points?: number
}

interface Project {
  id: string
  title: string
}

/**
 * Dashboard 客户端数据获取包装器
 * 将原本在 layout.tsx 中的异步数据库查询移到客户端，
 * 避免服务端 layout 异常导致整个路由段白屏。
 */
export default function DashboardShell({ children }: { chi
  ldren: React.ReactNode }) {
  const [user, setUser] = useState<User | null>(null)
  const [recentProjects, setRecentProjects] = useState<Pro
  ject[]>([])
  const [loading, setLoading] = useState(true)

  useEffect(() => {
    let mounted = true
    async function fetchData() {
      try {
        setLoading(true)

        // 并行获取用户信息和最近项目
        const [userData, projectsData] = await Promise.all
  ([
          apiClient<{ user: User }>('/api/user', { redirec
```

```
  tOnAuthError: false }),
        apiClient<{ projects: Project[] }>('/api/project
  s', { redirectOnAuthError: false }),
      ])

      if (!mounted) return

      setUser(userData?.user || null)
      setRecentProjects((projectsData?.projects || []).s
  lice(0, 5))
    } catch (e: any) {
      if (!mounted) return
      console.error('[DashboardShell] fetch error:', e)
      // 后台数据同步失败时保留当前页面；真实未登录由路
  由中间件处理。
    } finally {
      if (mounted) setLoading(false)
    }
  }

  fetchData()
  return () => {
    mounted = false
  }
}, [])

return (
  <div className="flex min-h-screen bg-stone-50">
    <AppSidebar
      user={user || { name: '加载中...', email: '' }}
      recentProjects={recentProjects}
      loading={loading}
    />
    <main className="flex flex-1 flex-col min-h-screen">
      <Header />
      <div className="flex-1 overflow-auto p-6">
        {loading && (
          <div className="mb-4 flex items-center gap-2 t
  ext-xs text-stone-400">
            <div className="h-3 w-3 animate-spin rounded
  -full border border-stone-300 border-t-amber-600" />
            正在同步侧边栏数据...
          </div>
        )}
        {children}
      </div>
    </main>
  </div>
)
}
```

```
/* ===== FILE: app/(dashboard)/assets/page.tsx ===== */
'use client'

import { use, useMemo, useState } from 'react'
import useSWR from 'swr'
import { Bookmark, LoaderCircle, Plus, RefreshCw, Sparkles
  , Upload } from 'lucide-react'
import { ASSET_LIBRARY_TEMPLATES } from '@/lib/asset-libra
  ry'
import { getImageGenerationCost } from '@/lib/points-confi
  g'
import { IMAGE_MODELS } from '@/lib/models-config'

type UserAsset = {
  id: string
  kind: 'CHARACTER' | 'ENVIRONMENT' | 'REFERENCE'
  source: 'UPLOAD' | 'GENERATED'
  title?: string | null
  description?: string | null
  url: string
  prompt?: string | null
  createdAt: string
}

const fetcher = (url: string) => fetch(url).then((r) => r.
  json())

export default function UserAssetsPage({
  searchParams,
}: {
  searchParams: Promise<{ projectId?: string }>
}) {
  const resolvedSearchParams = use(searchParams)
  const [kind, setKind] = useState<'ALL' | 'CHARACTER' | '
    ENVIRONMENT' | 'REFERENCE'>('ALL')
  const [uploading, setUploading] = useState(false)
  const [generating, setGenerating] = useState(false)
  const [message, setMessage] = useState('')
  const [selectedRefs, setSelectedRefs] = useState<Set<str
    ing>>(new Set())
  const [form, setForm] = useState({
    kind: 'CHARACTER' as 'CHARACTER' | 'ENVIRONMENT',
    templateId: ASSET_LIBRARY_TEMPLATES.find((t) => t.kind
     === 'CHARACTER')?.id || '',
    imageModel: IMAGE_MODELS.primary as string,
    title: '',
    description: '',
  })

  const apiUrl = kind === 'ALL' ? '/api/user-assets' : `/a
```

```
  pi/user-assets?kind=${kind}`
const { data, mutate, isLoading } = useSWR(apiUrl, fetch
  er)
const assets: UserAsset[] = data?.assets || []
const projectId = resolvedSearchParams.projectId || ''
const templates = useMemo(
  () => ASSET_LIBRARY_TEMPLATES.filter((t) => t.kind ===
   form.kind),
  [form.kind]
)

async function uploadAsset(file: File) {
  setUploading(true)
  setMessage('')
  try {
    const fd = new FormData()
    fd.append('file', file)
    fd.append('kind', kind === 'ALL' ? 'REFERENCE' : kin
  d)
    fd.append('title', file.name)
    const res = await fetch('/api/user-assets', { method
  : 'POST', body: fd })
    const json = await res.json().catch(() => ({}))
    if (!res.ok) throw new Error(json.message || json.er
  ror || `HTTP ${res.status}`)
    await mutate()
    setMessage('已上传到资产库')
  } catch (e: any) {
    setMessage(e?.message || '上传失败')
  } finally {
    setUploading(false)
  }
}

async function generateAsset() {
  setGenerating(true)
  setMessage('')
  try {
    const res = await fetch('/api/user-assets/generate',
   {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({
        ...form,
        referenceAssetIds: Array.from(selectedRefs),
      }),
    })
    const json = await res.json().catch(() => ({}))
    if (!res.ok) throw new Error(json.message || json.er
  ror || `HTTP ${res.status}`)
```

```
    await mutate()
    setMessage(`生成完成，已扣 ${json.cost ?? getImageGe
  nerationCost(form.imageModel)} 点`)
  } catch (e: any) {
    setMessage(e?.message || '生成失败')
  } finally {
    setGenerating(false)
  }
}

async function copyToProject(assetId: string) {
  if (!projectId) return
  setMessage('')
  const res = await fetch(`/api/user-assets/${assetId}/c
  opy-to-project`, {
    method: 'POST',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify({ projectId }),
  })
  const json = await res.json().catch(() => ({}))
  if (!res.ok) {
    setMessage(json.message || json.error || '加入项目失
  败')
    return
  }
  setMessage('已加入当前项目参考图')
}

function updateKind(nextKind: 'CHARACTER' | 'ENVIRONMENT
  ') {
  const template = ASSET_LIBRARY_TEMPLATES.find((t) => t
  .kind === nextKind)
  setForm((prev) => ({ ...prev, kind: nextKind, template
  Id: template?.id || '' }))
}

return (
  <div className="space-y-5">
    <div className="flex flex-wrap items-center justify-
  between gap-3">
      <div>
        <h1 className="text-xl font-bold text-stone-900"
  >资产库</h1>
        <p className="mt-1 text-sm text-stone-500">管理
  可跨项目复用的人物形象、环境图片和参考素材。</p>
      </div>
      {message && <div className="rounded-md bg-stone-10
  0 px-3 py-1.5 text-sm text-stone-700">{message}</div>}
    </div>

```

```
  <div className="grid gap-4 lg:grid-cols-[360px_1fr]"
>
    <div className="space-y-4">
      <div className="rounded-lg border border-stone-2
00 bg-white p-4">
        <h2 className="mb-3 text-sm font-semibold text
-stone-800">上传素材</h2>
        <label className="flex h-28 cursor-pointer fle
x-col items-center justify-center rounded-lg border bo
rder-dashed border-stone-300 bg-stone-50 text-sm text-
stone-500 hover:bg-stone-100">
          {uploading ? <LoaderCircle className="mb-2 h
-5 w-5 animate-spin" /> : <Upload className="mb-2 h-5
w-5" />}
          选择图片上传
          <input
            type="file"
            accept="image/png,image/jpeg,image/webp"
            className="hidden"
            disabled={uploading}
            onChange={(e) => {
              const file = e.target.files?.[0]
              if (file) uploadAsset(file)
              e.currentTarget.value = ''
            }}
          />
        </label>
      </div>

      <div className="rounded-lg border border-stone-2
00 bg-white p-4">
        <h2 className="mb-3 text-sm font-semibold text
-stone-800">生成资产</h2>
        <div className="mb-3 grid grid-cols-2 gap-2">
          <button
            onClick={() => updateKind('CHARACTER')}
            className={`rounded-md px-3 py-2 text-sm ${
form.kind === 'CHARACTER' ? 'bg-stone-900 text-white'
 : 'bg-stone-100 text-stone-700'}`}
          >
            人物形象
          </button>
          <button
            onClick={() => updateKind('ENVIRONMENT')}
            className={`rounded-md px-3 py-2 text-sm ${
form.kind === 'ENVIRONMENT' ? 'bg-stone-900 text-whit
e' : 'bg-stone-100 text-stone-700'}`}
          >
            环境图片
          </button>
```

```
401           </div>
402           <label className="text-xs text-stone-500">模板
403   </label>
404           <select
405             className="mt-1 w-full rounded-md border bor
406   der-stone-200 px-3 py-2 text-sm"
407             value={form.templateId}
408             onChange={(e) => setForm((prev) => ({ ...pre
409   v, templateId: e.target.value }))}
410           >
411             {templates.map((t) => (
412               <option key={t.id} value={t.id}>{t.name}</
413   option>
414             ))}
415           </select>
416           <label className="mt-3 block text-xs text-ston
417   e-500">生图模型</label>
418           <select
419             className="mt-1 w-full rounded-md border bor
420   der-stone-200 px-3 py-2 text-sm"
421             value={form.imageModel}
422             onChange={(e) => setForm((prev) => ({ ...pre
423   v, imageModel: e.target.value }))}
424           >
425             {IMAGE_MODELS.available.filter((m: any) => !
426   (m as any).disabled).map((m) => (
427               <option key={m.id} value={m.id}>{m.label}<
428   /option>
429             ))}
430           </select>
431           <label className="mt-3 block text-xs text-ston
432   e-500">名称</label>
433           <input
434             className="mt-1 w-full rounded-md border bor
435   der-stone-200 px-3 py-2 text-sm"
436             value={form.title}
437             onChange={(e) => setForm((prev) => ({ ...pre
438   v, title: e.target.value }))}
439           />
440           <label className="mt-3 block text-xs text-ston
441   e-500">描述</label>
442           <textarea
443             className="mt-1 min-h-28 w-full rounded-md b
444   order border-stone-200 px-3 py-2 text-sm"
445             value={form.description}
446             onChange={(e) => setForm((prev) => ({ ...pre
447   v, description: e.target.value }))}
448             placeholder="写清外貌、服装、场景、材质、情
449   绪或品牌特征"
450           />
```

```
        <button
          onClick={generateAsset}
          disabled={generating || !form.templateId}
          className="mt-3 flex w-full items-center jus
tify-center gap-2 rounded-md bg-stone-900 px-3 py-2 te
xt-sm font-medium text-white disabled:opacity-50"
        >
          {generating ? <LoaderCircle className="h-4 w
-4 animate-spin" /> : <Sparkles className="h-4 w-4" />
}
          生成并保存到资产库（{getImageGenerationCost(
form.imageModel)} 点）
        </button>
      </div>
    </div>

    <div className="rounded-lg border border-stone-200
 bg-white">
      <div className="flex flex-wrap items-center just
ify-between gap-2 border-b border-stone-200 p-4">
        <div className="flex gap-2">
          {(['ALL', 'CHARACTER', 'ENVIRONMENT', 'REFER
ENCE'] as const).map((k) => (
            <button
              key={k}
              onClick={() => setKind(k)}
              className={`rounded-md px-3 py-1.5 text-
sm ${kind === k ? 'bg-stone-900 text-white' : 'bg-ston
e-100 text-stone-600'}`}
            >
              {k === 'ALL' ? '全部' : k === 'CHARACTER
' ? '人物' : k === 'ENVIRONMENT' ? '环境' : '上传'}
            </button>
          ))}
        </div>
        <button onClick={() => mutate()} className="ro
unded-md bg-stone-100 p-2 text-stone-600">
          <RefreshCw className="h-4 w-4" />
        </button>
      </div>
      <div className="grid gap-3 p-4 sm:grid-cols-2 xl
:grid-cols-3">
        {isLoading ? (
          <div className="col-span-full flex h-40 item
s-center justify-center text-stone-400">
            <LoaderCircle className="h-5 w-5 animate-s
pin" />
          </div>
        ) : assets.length === 0 ? (
          <div className="col-span-full flex h-40 item
```

```
s-center justify-center text-sm text-stone-400">暂无资产</div>
        ) : assets.map((asset) => {
          const selected = selectedRefs.has(asset.id)
          return (
            <div key={asset.id} className="overflow-hidden rounded-lg border border-stone-200 bg-white">
              <div className="aspect-video bg-stone-100">
                <img src={asset.url} alt="" className="h-full w-full object-cover" />
              </div>
              <div className="space-y-2 p-3">
                <div className="flex items-start justify-between gap-2">
                  <div>
                    <p className="line-clamp-1 text-sm font-medium text-stone-800">{asset.title || asset.kind}</p>
                    <p className="text-xs text-stone-400">{asset.kind} · {asset.source}</p>
                  </div>
                  <button
                    onClick={() => {
                      setSelectedRefs((prev) => {
                        const next = new Set(prev)
                        selected ? next.delete(asset.id) : next.add(asset.id)
                        return next
                      })
                    }}
                    className={`rounded-md p-1.5 ${selected ? 'bg-emerald-100 text-emerald-700' : 'bg-stone-100 text-stone-500'}`}
                    title="作为生成参考图"
                  >
                    <Bookmark className="h-4 w-4" />
                  </button>
                </div>
                {projectId && (
                  <button
                    onClick={() => copyToProject(asset.id)}
                    className="flex w-full items-center justify-center gap-1.5 rounded-md border border-stone-200 px-2 py-1.5 text-xs font-medium text-stone-700 hover:bg-stone-50"
                  >
                    <Plus className="h-3.5 w-3.5" />
                    加入当前项目参考图
```

```
                    </button>
                  )}
                </div>
              </div>
            )
          })}
        </div>
      </div>
    </div>
  </div>
)
}
/* ===== FILE: app/(dashboard)/dashboard/_components/Proje
  ctCard.tsx ===== */
'use client'

import { useState } from 'react'
import Link from 'next/link'
import { Clock, Trash2, LoaderCircle } from 'lucide-react'
import { apiClient } from '@/lib/api-client'

interface ProjectCardProps {
  project: {
    id: string
    title: string
    rawIdea: string
    createdAt: string
    assetCount: number
    progressText: string
  }
  onDeleted: (id: string) => void
  onError?: (message: string) => void
}

export default function ProjectCard({ project, onDeleted,
  onError }: ProjectCardProps) {
  const [isHovered, setIsHovered] = useState(false)
  const [isDeleting, setIsDeleting] = useState(false)

  const handleDelete = async () => {
    const confirmed = window.confirm(
      `确定要删除项目「${project.title}」吗？此操作不可撤
  销，将删除该项目下的所有资产文件。`
    )
    if (!confirmed) return

    setIsDeleting(true)
    try {
      const data = await apiClient<{ success?: boolean; me
  ssage?: string }>(`/api/projects/${project.id}`, { met
```

```
  hod: 'DELETE' })
    if (!data.success) {
      throw new Error(data.message || '删除失败')
    }
    onDeleted(project.id)
  } catch (err: any) {
    console.error('[PROJECT-DELETE] 失败:', err)
    onError?.(err?.message || '删除失败')
  } finally {
    setIsDeleting(false)
  }
}

return (
  <div
    className="group relative rounded-lg border border-s
  tone-200 bg-white p-5 shadow-sm transition hover:borde
  r-stone-300 hover:shadow-md"
    onMouseEnter={() => setIsHovered(true)}
    onMouseLeave={() => setIsHovered(false)}
  >
    {/* 删除按钮 - 悬停时显示 */}
    <div
      className={`absolute top-2 right-2 z-10 transition
  -opacity duration-200 ${
        isHovered && !isDeleting ? 'opacity-100' : 'opac
  ity-0'
      }`}
    >
      <button
        onClick={(e) => {
          e.preventDefault()
          e.stopPropagation()
          handleDelete()
        }}
        disabled={isDeleting}
        className="flex items-center gap-1 rounded-md bg
  -red-50 px-2 py-1 text-xs font-medium text-red-600 tra
  nsition-colors hover:bg-red-100 disabled:opacity-50"
        title="删除项目"
      >
        <Trash2 className="h-3 w-3" />
        删除
      </button>
    </div>

    {/* 删除中遮罩 */}
    {isDeleting && (
      <div className="absolute inset-0 z-20 flex items-c
  enter justify-center rounded-lg bg-white/80">
```

```
        <div className="flex items-center gap-2 text-sm
text-stone-500">
          <LoaderCircle className="h-4 w-4 animate-spin"
 />
          删除中...
        </div>
      </div>
    )}

    <Link href={`/project/${project.id}`} className="blo
ck">
      <div className="flex items-start justify-between">
        <h3 className="font-semibold text-stone-800 grou
p-hover:text-stone-900 pr-16">
          {project.title}
        </h3>
        <span className="rounded-full bg-stone-100 px-2
py-0.5 text-[10px] font-medium text-stone-500 shrink-0
">
          {project.assetCount} 资产
        </span>
      </div>
      <p className="mt-2 line-clamp-2 text-sm text-stone
-500">{project.rawIdea}</p>
      <div className="mt-4 flex items-center gap-3 text-
xs text-stone-400">
        <span className="inline-flex items-center gap-1"
>
          <Clock className="h-3 w-3" />
          {new Date(project.createdAt).toLocaleDateStrin
g('zh-CN')}
        </span>
        <span className="rounded-md bg-amber-50 px-1.5 p
y-0.5 text-amber-700">
          {project.progressText}
        </span>
      </div>
    </Link>
  </div>
)
}
/* ===== FILE: app/(dashboard)/dashboard/_components/Proje
   ctList.tsx ===== */
'use client'

import { useState, useCallback } from 'react'
import ProjectCard from './ProjectCard'

interface Project {
  id: string
```

```
  title: string
  rawIdea: string
  createdAt: string
  assetCount: number
  progressText: string
}

interface ProjectListProps {
  initialProjects: Project[]
}

export default function ProjectList({ initialProjects }: P
    rojectListProps) {
  const [projects, setProjects] = useState<Project[]>(init
    ialProjects)
  const [toast, setToast] = useState<{ kind: 'success' | '
    error'; message: string } | null>(null)

  const handleDeleted = useCallback((id: string) => {
    setProjects((prev) => prev.filter((p) => p.id !== id))
    setToast({ kind: 'success', message: '项目已删除' })
    setTimeout(() => setToast(null), 3000)
  }, [])

  const handleError = useCallback((message: string) => {
    setToast({ kind: 'error', message })
    setTimeout(() => setToast(null), 5000)
  }, [])

  if (projects.length === 0) {
    return (
      <div className="flex flex-col items-center justify-c
    enter rounded-xl border border-dashed border-stone-300
     bg-white py-16 text-center">
        {/* 空状态与原始页面一致 */}
      </div>
    )
  }

  return (
    <>
      {toast && (
        <div
          className={`fixed bottom-4 right-4 z-50 rounded-
    lg px-4 py-2 text-sm font-medium shadow-lg transition-
    all ${
            toast.kind === 'success'
              ? 'bg-emerald-50 text-emerald-700 border bor
    der-emerald-200'
              : 'bg-red-50 text-red-700 border border-red-
```

```
    200'
          }`}
        >
          {toast.message}
        </div>
      )}
      <div className="grid grid-cols-1 gap-4 md:grid-cols-
    2 xl:grid-cols-3">
        {projects.map((project) => (
          <ProjectCard
            key={project.id}
            project={project}
            onDeleted={handleDeleted}
            onError={handleError}
          />
        ))}
      </div>
    </>
  )
}
/* ===== FILE: app/(dashboard)/dashboard/page.tsx ===== */
export const dynamic = 'force-dynamic'

import { redirect } from 'next/navigation'
import { prisma } from '@/lib/prisma'
import { projectDashboardSelect } from '@/lib/db/project-s
    elect'
import { WORKFLOW_STEPS } from '@/lib/workflow'
import { getCurrentUser } from '@/lib/auth-helpers'
import Link from 'next/link'
import {
  PlusCircle,
  FolderOpen,
  ImageIcon,
  Video,
  BarChart3,
  Clock,
  Film,
} from 'lucide-react'
import ProjectList from './_components/ProjectList'

function getStartOfWeek(date: Date): Date {
  const d = new Date(date)
  const day = d.getDay()
  const diff = d.getDate() - day
  d.setDate(diff)
  d.setHours(0, 0, 0, 0)
  return d
}

```

```
function getProgressText(steps?: { order?: number | null;
  status?: string | null }[] | null): string {
 if (!steps || steps.length === 0) return '刚创建'
 const validOrders = steps
  .filter((s) => s && (s.status === 'COMPLETED' || s.sta
  tus === 'SKIPPED'))
  .map((s) => s.order)
  .filter((o): o is number => typeof o === 'number' && !
  isNaN(o))
 const maxCompleted = validOrders.length > 0 ? Math.max(-
  1, ...validOrders) : -1
 const currentOrder = maxCompleted + 1
 if (currentOrder >= WORKFLOW_STEPS.length) return '已完
  成'
 const step = WORKFLOW_STEPS[currentOrder]
 if (!step) return '进行中'
 return `第 ${currentOrder + 1} 步：${step.label}`
}

export default async function DashboardPage() {
 try {
  const user = await getCurrentUser()
  if (!user) redirect('/login')

  const weekAgo = new Date(Date.now() - 7 * 24 * 60 * 60
   * 1000)

  const userId = user.id
  const [activeProjectsCount, weeklyAssetsCount, videoAs
  setsCount, userAssets] =
    await Promise.all([
      prisma.project.count({ where: { status: 'ACTIVE',
  userId } }),
      prisma.asset.count({
        where: {
          project: { userId },
          createdAt: { gte: weekAgo },
        },
      }),
      prisma.asset.count({
        where: {
          project: { userId },
          type: 'VIDEO',
        },
      }),
      prisma.asset.findMany({
        where: { project: { userId } },
        select: { id: true, projectId: true },
      }),
    ])
```

```
851
852   const evaluationsCount = userAssets.length > 0
853     ? await prisma.evaluation.count({
854         where: { assetId: { in: userAssets.map((a: { id:
855    string }) => a.id) } },
856       })
857     : 0
858
859   const projects = await prisma.project.findMany({
860     where: { status: 'ACTIVE', userId },
861     orderBy: { updatedAt: 'desc' },
862     take: 20,
863     select: projectDashboardSelect,
864   })
865
866   const stats = [
867     { label: '进行中的项目', value: activeProjectsCount,
868    icon: FolderOpen, color: 'text-blue-600', bg: 'bg-blu
869   e-50' },
870     { label: '本周生成资产', value: weeklyAssetsCount, i
871   con: ImageIcon, color: 'text-emerald-600', bg: 'bg-eme
872   rald-50' },
873     { label: '已完成视频片段', value: videoAssetsCount,
874   icon: Video, color: 'text-violet-600', bg: 'bg-violet-
875   50' },
876     { label: '最近评测次数', value: evaluationsCount, ic
877   on: BarChart3, color: 'text-amber-600', bg: 'bg-amber-
878   50' },
879   ]
880
881   return (
882     <div className="space-y-8">
883       {/* 顶部欢迎 + 新建按钮 */}
884       <div className="flex flex-col gap-4 sm:flex-row sm
885   :items-center sm:justify-between">
886         <div>
887           <h1 className="text-2xl font-bold tracking-tig
888   ht text-stone-800">
889             欢迎回来，{user.name || '创作者'}
890           </h1>
891           <p className="mt-1 text-sm text-stone-500">管
892   理你的 AI 影视创作项目</p>
893         </div>
894         <Link
895           href="/project/new"
896           className="inline-flex items-center gap-2 roun
897   ded-lg bg-stone-900 px-5 py-2.5 text-sm font-medium te
898   xt-white shadow-sm transition hover:bg-stone-800"
899         >
900           <PlusCircle className="h-4 w-4" />
```

```
          新建项目
        </Link>
      </div>

      {/* 统计卡片 */}
      <div className="grid grid-cols-2 gap-4 lg:grid-col
s-4">
        {stats.map((s) => {
          const Icon = s.icon
          return (
            <div
              key={s.label}
              className="rounded-lg border border-stone-
200 bg-white p-4 shadow-sm"
            >
              <div className="flex items-center gap-3">
                <div className={`flex h-9 w-9 items-cent
er justify-center rounded-lg ${s.bg}`}>
                  <Icon className={`h-4 w-4 ${s.color}`}
 />
                </div>
                <div>
                  <p className="text-xs text-stone-500">
{s.label}</p>
                  <p className="text-xl font-bold text-s
tone-800">{s.value}</p>
                </div>
              </div>
            </div>
          )
        })}
      </div>

      {/* 项目列表 */}
      <div>
        <h2 className="mb-4 text-sm font-semibold upperc
ase tracking-wider text-stone-500">
          最近项目
        </h2>

        {projects.length === 0 ? (
          <div className="flex flex-col items-center jus
tify-center rounded-xl border border-dashed border-sto
ne-300 bg-white py-16 text-center">
            <div className="flex h-14 w-14 items-center
justify-center rounded-full bg-stone-100">
              <Film className="h-7 w-7 text-stone-400" /
>
            </div>
            <h3 className="mt-4 text-base font-semibold
```

```
951   text-stone-700">
952                 开始你的第一个 AI 影视项目
953               </h3>
954               <p className="mt-1 max-w-sm text-sm text-sto
955   ne-500">
956                 输入灵感，让 AI 帮你完成从创意到成片的全流
957   程创作
958               </p>
959               <Link
960                 href="/project/new"
961                 className="mt-5 inline-flex items-center g
962   ap-2 rounded-lg bg-stone-900 px-4 py-2 text-sm font-me
963   dium text-white transition hover:bg-stone-800"
964               >
965                 <PlusCircle className="h-4 w-4" />
966                 新建项目
967               </Link>
968             </div>
969           ) : (
970             <ProjectList
971               initialProjects={projects.map((p: any) => ({
972                 id: p.id,
973                 title: p.title || '未命名项目',
974                 rawIdea: p.rawIdea || '',
975                 createdAt: p.createdAt ? new Date(p.create
976   dAt).toISOString() : new Date().toISOString(),
977                 assetCount: p._count?.assets ?? 0,
978                 progressText: getProgressText(p.steps),
979               }))}
980             />
981           )}
982         </div>
983       </div>
984     )
985   } catch (e: any) {
986     // [DASHBOARD-DEBUG] 将服务器端错误输出到日志，便于排
987     查
988     console.error('[DASHBOARD] Server Component 渲染错误:'
989     , e?.message || e)
990     console.error('[DASHBOARD] 错误堆栈:', e?.stack || '无
991     堆栈')
992     throw e
993   }
994 }
995 /* ===== FILE: app/(dashboard)/error.tsx ===== */
996 'use client'
997
998 import { useEffect } from 'react'
999
1000 /**
```

```
1001 * Dashboard 段错误边界
1002 * 捕获仪表盘及子页面（项目、资产库等）中的渲染错误
1003 */
1004export default function DashboardError({
1005  error,
1006  reset,
1007}: {
1008  error: Error & { digest?: string }
1009  reset: () => void
1010}) {
1011  useEffect(() => {
1012    console.error('[Dashboard Error]', error)
1013  }, [error])
1014
1015  return (
1016    <div className="flex min-h-screen flex-col items-cente
1017    r justify-center bg-stone-50 p-6">
1018      <h2 className="mb-2 text-xl font-semibold text-stone
1019    -800">页面加载出错</h2>
1020      <p className="mb-4 max-w-md text-center text-sm text
1021    -stone-500">
1022        {error.message || '仪表盘页面加载失败，请刷新重试'
1023    }
1024      </p>
1025      <div className="flex gap-3">
1026        <button
1027          onClick={() => reset()}
1028          className="rounded-lg bg-stone-900 px-4 py-2 tex
1029    t-sm font-medium text-white hover:bg-stone-800"
1030        >
1031          重试
1032        </button>
1033        <a
1034          href="/dashboard"
1035          className="rounded-lg border border-stone-200 bg
1036    -white px-4 py-2 text-sm font-medium text-stone-600 ho
1037    ver:bg-stone-50"
1038        >
1039          返回仪表盘
1040        </a>
1041      </div>
1042    </div>
1043  )
1044}
1045/* ===== FILE: app/(dashboard)/groups/[id]/ContributionPan
1046    el.tsx ===== */
1047'use client'
1048
1049import { useEffect, useState } from 'react'
1050import { Loader2 } from 'lucide-react'
```

```
import { apiClient } from '@/lib/api-client'

type ContributionData = {
  summary: {
    requestCount: number
    providerCallCount: number
    /** 资产总数（含被替换的历史版本） */
    outputCount: number
    /** 去重后的业务目标数（项目/步骤/镜头或风格/角色/帧角
    色） */
    distinctTargetCount: number
    /** 当前有效贡献：去重目标里仍被项目采用的数量 */
    adoptedCount: number
    adoptionRate: number
    netPointsCost: number
    providerCost: number | null
    currency: string | null
    unreconciledAttemptCount: number
  }
  members: Array<{
    user: { id: string; name: string | null; email: string
     }
    requestCount: number
    outputCount: number
    distinctTargetCount: number
    adoptedCount: number
    adoptionRate: number
    netPointsCost: number
  }>
  projects: Array<{ id: string; title: string }>
  truncatedSummary?: boolean
  maxScan?: number
  rows: Array<{
    id: string
    actionKey: string
    category: string
    status: string
    member: { id: string; name: string | null; email: stri
    ng }
    project: { id: string; title: string } | null
    stepName: string | null
    scopeKey: string | null
    models: string[]
    requestCount: number
    providerCallCount: number
    outputCount: number
    netPointsCost: number
    providerCost: number | null
    currency: string | null
    startedAt: string
```

```
    durationMs: number | null
    results: Array<{ adoptionStatus: string; shotId: strin
    g | null; actNumber: number | null }>
  }>
}

const statusLabel: Record<string, string> = {
  SUCCEEDED: '成功', PARTIAL: '部分成功', FAILED: '失败',
    RUNNING: '进行中', SUBMITTED: '已提交', CANCELLED: '已
    取消',
  IMPORTED: '已导入',
}

export default function ContributionPanel({ groupId }: { g
    roupId: string }) {
  const [data, setData] = useState<ContributionData | null
    >(null)
  const [loading, setLoading] = useState(true)
  const [error, setError] = useState<string | null>(null)
  const [days, setDays] = useState(30)
  const [projectId, setProjectId] = useState('')

  useEffect(() => {
    let active = true
    setLoading(true)
    setError(null)
    const query = new URLSearchParams({ days: String(days)
    , pageSize: '50' })
    if (projectId) query.set('projectId', projectId)
    apiClient<ContributionData>(`/api/groups/${groupId}/co
    ntributions?${query}`)
      .then((result) => { if (active) setData(result) })
      .catch((err: unknown) => { if (active) setError(err
    instanceof Error ? err.message : '贡献数据加载失败') }
    )
      .finally(() => { if (active) setLoading(false) })
    return () => { active = false }
  }, [days, groupId, projectId])

  if (loading && !data) return <div className="flex justif
    y-center py-16"><Loader2 className="h-6 w-6 animate-sp
    in text-stone-400" /></div>
  if (error) return <div className="rounded-lg border bord
    er-red-200 bg-red-50 p-4 text-sm text-red-700">{error}
    </div>
  if (!data) return null

  const cards = [
    ['生成请求', data.summary.requestCount],
    ['供应商调用', data.summary.providerCallCount],
```

```
  ['产出资产总数', data.summary.outputCount],
  ['去重目标数', data.summary.distinctTargetCount],
  ['当前有效贡献', data.summary.adoptedCount],
  ['当前采用率', `${(data.summary.adoptionRate * 100).toFixed(1)}%`],
  ['净消耗点数', data.summary.netPointsCost],
]

return (
  <div className="space-y-5">
    <div className="flex flex-wrap items-center justify-between gap-3">
      <div className="min-w-0 flex-1">
        <p className="text-sm text-stone-600">
          <strong className="font-semibold text-stone-800">三个数字的区别：</strong>
          「产出资产总数」是数据库里的资产行数，
          <em className="not-italic">同一个镜头重生成几次就会 +几</em>；
          「去重目标数」按项目 / 步骤 / 镜头（风格、角色、帧角色同理）合并，
          代表<em className="not-italic">实际动过多少个目标</em>；
          「当前有效贡献」只统计去重后<em className="not-italic">仍被项目采用</em>的目标。
        </p>
        <p className="mt-1 text-xs text-stone-500">
          采用率 = 有效贡献 ÷ 去重目标数。无法追溯操作者的旧数据标记为“未知”，不归给项目所有者。
        </p>
        {data.truncatedSummary && (
          <p className="mt-2 rounded border border-amber-200 bg-amber-50 px-2 py-1 text-xs text-amber-800">
            数据量超过单次扫描上限（{data.maxScan} 条/表），本次统计已截断。请缩小时间范围或分项目查看。
          </p>
        )}
      </div>
      <div className="flex gap-2">
        <a href={`/api/groups/${groupId}/contributions/export?days=${days}${projectId ? `&projectId=${encodeURIComponent(projectId)}` : ''}`} className="rounded-lg border border-stone-200 bg-white px-3 py-2 text-sm text-stone-600 hover:bg-stone-50">导出 CSV</a>
        <select value={projectId} onChange={(event) => setProjectId(event.target.value)} className="rounded-lg border border-stone-200 bg-white px-3 py-2 text-sm">
          <option value="">全部项目</option>
          {data.projects.map((project) => <option key={project.id} value={project.id}>{project.title}</option>
```

```
)}
      </select>
      <select value={days} onChange={(event) => setDay
s(Number(event.target.value))} className="rounded-lg b
order border-stone-200 bg-white px-3 py-2 text-sm">
        <option value={7}>近 7 天</option><option valu
e={30}>近 30 天</option><option value={90}>近 90 天</o
ption><option value={365}>近一年</option>
      </select>
    </div>
  </div>

  <div className="grid gap-3 sm:grid-cols-2 lg:grid-co
ls-4">
    {cards.map(([label, value]) => <div key={label} cl
assName="rounded-lg border border-stone-200 bg-white p
-4"><p className="text-xs text-stone-500">{label}</p><
p className="mt-1 text-xl font-semibold text-stone-800
">{value}</p></div>)}
  </div>

  <div className="overflow-hidden rounded-lg border bo
rder-stone-200 bg-white">
    <div className="border-b px-4 py-3 text-sm font-me
dium text-stone-700">成员汇总</div>
    <div className="overflow-x-auto">
      <table className="w-full min-w-[860px] text-left
 text-sm">
        <thead className="bg-stone-50 text-xs text-sto
ne-600"><tr>{['成员', '生成请求', '资产总数', '去重目
标数', '有效贡献', '采用率', '净点数'].map((item) => <
th key={item} className="px-4 py-3 font-medium">{item}
</th>)}</tr></thead>
        <tbody className="divide-y">{data.members.map(
(member) => <tr key={member.user.id}><td className="px
-4 py-3"><div>{member.user.name || '未命名'}</div><div
 className="text-xs text-stone-500">{member.user.email
}</div></td><td className="px-4 py-3">{member.requestC
ount}</td><td className="px-4 py-3 text-stone-500">{me
mber.outputCount}</td><td className="px-4 py-3 font-me
dium">{member.distinctTargetCount}</td><td className="
px-4 py-3">{member.adoptedCount}</td><td className="px
-4 py-3">{(member.adoptionRate * 100).toFixed(1)}%</td
><td className="px-4 py-3">{member.netPointsCost}</td>
</tr>)}</tbody>
      </table>
    </div>
  </div>

  <div className="overflow-hidden rounded-lg border bo
```

```
rder-stone-200 bg-white">
    <div className="border-b px-4 py-3 text-sm font-me
dium text-stone-700">生成明细</div>
    <div className="overflow-x-auto">
      <table className="w-full min-w-[1120px] text-lef
t text-sm">
        <thead className="bg-stone-50 text-xs text-sto
ne-500"><tr>{['时间', '成员', '项目', '任务 / 镜头', '
模型', '次数', '耗时', '成本', '采用状态', '状态'].map
((item) => <th key={item} className="px-4 py-3 font-me
dium">{item}</th>)}</tr></thead>
        <tbody className="divide-y">{data.rows.map((ro
w) => {
          const firstResult = row.results[0]
          const target = firstResult?.shotId ? `第${fi
rstResult.actNumber || '-'}幕 / ${firstResult.shotId}`
 : row.scopeKey || row.stepName || row.actionKey
          const adopted = row.results.some((result) =>
 result.adoptionStatus === 'ADOPTED' || result.adoptio
nStatus === 'ACTIVE')
          const decisionLabel = adopted ? '已采用' : '
未采用'
          return <tr key={row.id} className="hover:bg-
stone-50"><td className="whitespace-nowrap px-4 py-3 t
ext-xs text-stone-500">{new Date(row.startedAt).toLoca
leString('zh-CN')}</td><td className="px-4 py-3">{row.
member.name || row.member.email}</td><td className="px
-4 py-3">{row.project?.title || '-'}</td><td classname
="px-4 py-3"><div>{row.stepName || row.category}</div>
<div className="text-xs text-stone-400">{target}</div>
</td><td className="px-4 py-3 text-xs">{row.models.joi
n(', ') || '-'}</td><td className="px-4 py-3 text-xs">
请求 {row.requestCount} / 调用 {row.providerCallCount}
 / 结果 {row.outputCount}</td><td className="px-4 py-3
">{row.durationMs == null ? '-' : `${(row.durationMs /
 1000).toFixed(1)}s`}</td><td className="px-4 py-3"><d
iv>{row.netPointsCost} 点</div>{row.providerCost != nu
ll && <div className="text-xs text-stone-400">{row.pro
viderCost.toFixed(4)} {row.currency}</div>}</td><td cl
assName="px-4 py-3"><span className={`rounded-full px-
2 py-1 text-xs ${adopted ? 'bg-emerald-50 text-emerald
-700' : 'bg-stone-100 text-stone-600'}`}>{decisionLabe
l}</span></td><td className="px-4 py-3">{statusLabel[r
ow.status] || row.status}</td></tr>
        })}</tbody>
      </table>
    </div>
  </div>
</div>
)
```

```
}
/* ===== FILE: app/(dashboard)/groups/[id]/page.tsx =====
   */
'use client'

import { use, useEffect, useState } from 'react'
import { useRouter } from 'next/navigation'
import Link from 'next/link'
import {
  Users,
  FolderOpen,
  ArrowLeft,
  PlusCircle,
  Loader2,
  Trash2,
  Settings,
  Coins,
  LogOut,
  Mail,
  UserPlus,
  BarChart3,
} from 'lucide-react'
import { apiClient } from '@/lib/api-client'
import ContributionPanel from './ContributionPanel'

interface GroupDetail {
  id: string
  name: string
  description: string | null
  createdById: string
  points: number
  costMode: 'MEMBER_PAY' | 'GROUP_POOL'
  createdBy: { id: string; name: string | null; email: str
    ing }
  members: {
    id: string
    role: string
    user: { id: string; name: string | null; email: string
    ; image: string | null }
  }[]
  projects: {
    id: string
    title: string
    userId: string
    createdAt: string
    updatedAt: string
    _count: { assets: number }
  }[]
  _count: { members: number; projects: number }
}
```

```
1351
1352export default function GroupDetailPage({ params }: { para
1353    ms: Promise<{ id: string }> }) {
1354  const router = useRouter()
1355  const groupId = use(params).id
1356  const [group, setGroup] = useState<GroupDetail | null>(n
1357    ull)
1358  const [currentRole, setCurrentRole] = useState<string |
1359    null>(null)
1360  const [loading, setLoading] = useState(true)
1361  const [error, setError] = useState<string | null>(null)
1362  const [activeTab, setActiveTab] = useState<'projects' |
1363    'members' | 'contributions' | 'settings'>('projects')
1364
1365  const [inviteEmail, setInviteEmail] = useState('')
1366  const [inviteLoading, setInviteLoading] = useState(false
1367    )
1368  const [rechargeAmount, setRechargeAmount] = useState('')
1369  const [rechargeLoading, setRechargeLoading] = useState(f
1370    alse)
1371  const [deleteLoading, setDeleteLoading] = useState(false
1372    )
1373
1374  const isAdmin = currentRole === 'ADMIN'
1375
1376  async function fetchGroup() {
1377    try {
1378      const data = await apiClient<{
1379        group?: GroupDetail
1380        currentMembership?: { role: string; status: string
1381     }
1382        error?: string
1383        message?: string
1384      }>(`/api/groups/${groupId}`)
1385      if (data.group) {
1386        setGroup(data.group)
1387        setCurrentRole(data.currentMembership?.role || nul
1388    l)
1389      } else {
1390        setError(data.message || '加载失败')
1391      }
1392    } catch (err: any) {
1393      setError(err?.message || '网络错误')
1394    } finally {
1395      setLoading(false)
1396    }
1397  }
1398
1399  useEffect(() => {
1400    fetchGroup()
```

```
}, [groupId])

async function handleInvite(e: React.FormEvent) {
  e.preventDefault()
  if (!inviteEmail.trim()) return
  setInviteLoading(true)
  try {
    const endpoint = isAdmin ? `/api/groups/${groupId}/m
  embers` : `/api/groups/${groupId}/invitations`
    await apiClient(endpoint, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ email: inviteEmail.trim() }
  ),
    })
    setInviteEmail('')
    await fetchGroup()
  } catch (err: any) {
    setError(err?.message || '邀请失败')
  } finally {
    setInviteLoading(false)
  }
}

async function handleRemoveMember(userId: string) {
  if (!confirm('确定移除该成员吗？')) return
  try {
    await apiClient(`/api/groups/${groupId}/members/${us
  erId}`, { method: 'DELETE' })
    await fetchGroup()
  } catch (err: any) {
    setError(err?.message || '移除失败')
  }
}

async function handleRecharge(e: React.FormEvent) {
  e.preventDefault()
  const amount = parseInt(rechargeAmount, 10)
  if (!amount || amount <= 0) return
  setRechargeLoading(true)
  try {
    await apiClient(`/api/groups/${groupId}/recharge`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ amount }),
    })
    setRechargeAmount('')
    await fetchGroup()
  } catch (err: any) {
    setError(err?.message || '充值失败')
```

```
  } finally {
    setRechargeLoading(false)
  }
}

async function handleDelete() {
  if (!confirm('确定解散小组吗？小组内项目将变为个人项目
  ，不会删除。')) return
  setDeleteLoading(true)
  try {
    await apiClient(`/api/groups/${groupId}`, { method:
  'DELETE' })
    router.push('/groups')
  } catch (err: any) {
    setError(err?.message || '解散失败')
    setDeleteLoading(false)
  }
}

if (loading) {
  return (
    <div className="flex items-center justify-center py-
  20">
      <Loader2 className="h-8 w-8 animate-spin text-ston
  e-400" />
    </div>
  )
}

if (!group) {
  return (
    <div className="py-20 text-center text-stone-500">
      {error || '小组不存在'}
    </div>
  )
}

return (
  <div className="space-y-6">
    <div className="flex flex-col gap-4 sm:flex-row sm:i
  tems-center sm:justify-between">
      <div>
        <Link
          href="/groups"
          className="mb-2 inline-flex items-center gap-1
   text-sm text-stone-500 transition hover:text-stone-70
  0"
        >
          <ArrowLeft className="h-4 w-4" /> 返回小组列表
        </Link>
```

```
  const result: Record<StepId, ReturnType<typeof getStepDi
    splayState>> = {} as any
  for (const stepId of STEP_ORDER) {
    result[stepId] = getStepDisplayState(stepId, project)
  }
  return result
}

/**
 * 检查项目是否已完成某步骤（基于 WorkflowStep status = CO
    MPLETED 的兼容模式）
 * 用于在现有 WorkflowStep 数据之上计算 ProjectState
 */
export function computeProjectStateFromSteps(
  steps: Array<{ stepType: string; status: string; outputD
    ata?: unknown }>
): ProjectState {
  const isDone = (type: string) =>
    steps.some((s) => s.stepType === type && (s.status ===
     'COMPLETED' || s.status === 'SKIPPED'))
  const storyboardStep = steps.find((step) => step.stepTyp
    e === 'STORYBOARD')
  const storyboardOutput = storyboardStep?.outputData && t
    ypeof storyboardStep.outputData === 'object'
    ? storyboardStep.outputData as { shots?: Array<{ first
    FrameUrl?: unknown }> }
    : null
  const hasStoryboardFirstFrame = Array.isArray(storyboard
    Output?.shots)
    && storyboardOutput.shots.some((shot) => typeof shot?.
    firstFrameUrl === 'string' && shot.firstFrameUrl.trim(
    ).length > 0)

  return {
    stepIdeaDone: isDone('IDEATION'),
    stepFrameworkDone: isDone('FRAMEWORK'),
    stepStyleDone: isDone('STYLE'),
    stepCharacterDone: isDone('CHARACTER'),
    stepConceptDone: isDone('CONCEPT'),
    stepStoryboardDone: isDone('STORYBOARD'),
    stepStoryboardFirstframeDone: hasStoryboardFirstFrame,
    stepTrailerDone: isDone('TRAILER'),
    stepEndingDone: isDone('KEYFRAMES'),
    stepDirectDone: isDone('VIDEO_DIRECT'),
  }
}

/**
 * 根据 StepId 获取对应的 Prisma stepType 值
 */
```

```
export function stepIdToPrismaType(stepId: StepId): string {
  return STEP_ID_TO_TYPE[stepId]
}

/**
 * 根据 Prisma stepType 获取 StepId（可能返回 undefined，因为不是所有 12 步都在 DAG 中）
 */
export function prismaTypeToStepId(stepType: string): StepId | undefined {
  return TYPE_TO_STEP_ID[stepType]
}
/* ===== FILE: lib/workflow.ts ===== */
import { WorkflowStepType as PrismaStepType } from '@prisma/client';

// ========================================================
// 1. 步骤元数据定义（顺序、名称、图标、描述）
// ========================================================
export const WORKFLOW_STEPS = [
  { type: 'IDEATION' as const,     label: '创意扩散',   order: 0,  description: '基于元构思联想扩展',       icon: 'Lightbulb' },
  { type: 'FRAMEWORK' as const,    label: '框架搭建',   order: 1,  description: '设计背景、风格、角色、梗概', icon: 'Layout' },
  { type: 'STYLE' as const,        label: '风格统一',   order: 2,  description: '探索并确定统一视觉风格',    icon: 'Palette' },
  { type: 'CHARACTER' as const,    label: '人物设计',   order: 3,  description: '生成角色人设与概念图',      icon: 'User' },
  { type: 'CONCEPT' as const,      label: '概念图',     order: 4,  description: '每幕生成代表性场景',        icon: 'Image' },
  { type: 'TRAILER' as const,      label: '宣传片',     order: 5,  description: '30s 先导样例参考片',        icon: 'Play' },
  { type: 'STORYBOARD' as const,   label: '分镜设计',   order: 6,  description: '绘制分镜草图',              icon: 'Grid' },
  { type: 'KEYFRAMES' as const,    label: '生成尾帧',   order: 7,  description: '基于分镜起始帧生成尾帧',    icon: 'Frame' },
  { type: 'VIDEO_DIRECT' as const, label: '直生视频',   order: 8,  description: '基于起始帧/尾帧生成视频',
    icon: 'Video' },
```

```
  { type: 'VIDEO_RENDER' as const, label: 'AI 渲染',     or
    der: 9,  description: '实拍视频风格化（大动作）',   ic
    on: 'Wand' },
  { type: 'CAMERA' as const,       label: '电脑运镜',   or
    der: 10, description: '预设镜头运动+AI 渲染',       ico
    n: 'Camera' },
  { type: 'REVIEW' as const,       label: '评测优化',   or
    der: 11, description: '一致性检测与情感审核',       ic
    on: 'CheckCircle' },
] as const;

export type WorkflowStepType = typeof WORKFLOW_STEPS[numbe
    r]['type'];

// ========================================================
    =====
// 2. 可见步骤配置（前端工作流看板只渲染这 9 步）
// ========================================================
    =====
export const VISIBLE_STEP_TYPES = [
  'IDEATION',
  'FRAMEWORK',
  'STYLE',
  'CHARACTER',
  'CONCEPT',
  'TRAILER',
  'STORYBOARD',
  'KEYFRAMES',
  'VIDEO_DIRECT',
] as const;

export type VisibleStepType = typeof VISIBLE_STEP_TYPES[nu
    mber];

// 日志：确认可见步骤顺序
console.log('[WORKFLOW-REORG] 可见步骤:', JSON.stringify(V
    ISIBLE_STEP_TYPES));

// ========================================================
    =====
// 3. 状态流转校验函数
// ========================================================
    =====

/** 获取下一步 */
export function getNextStep(current: WorkflowStepType): Wo
    rkflowStepType | null {
  const idx = WORKFLOW_STEPS.findIndex(s => s.type === cur
    rent);
  return idx >= 0 && idx < WORKFLOW_STEPS.length - 1 ? WOR
```

```
1651    KFLOW_STEPS[idx + 1].type : null;
1652}
1653
1654/** 获取上一步 */
1655export function getPrevStep(current: WorkflowStepType): Wo
1656    rkflowStepType | null {
1657  const idx = WORKFLOW_STEPS.findIndex(s => s.type === cur
1658    rent);
1659  return idx > 0 ? WORKFLOW_STEPS[idx - 1].type : null;
1660}
1661
1662/** 判断是否可以跳转到目标步骤（目标在当前之后） */
1663export function canSkipTo(target: WorkflowStepType, curren
1664    t: WorkflowStepType): boolean {
1665  const targetIdx = WORKFLOW_STEPS.findIndex(s => s.type =
1666    == target);
1667  const currentIdx = WORKFLOW_STEPS.findIndex(s => s.type
1668    === current);
1669  // 允许向前跳过（但不可跳过未解锁的步骤，具体规则由业务
1670    层控制）
1671  return targetIdx > currentIdx;
1672}
1673
1674/** 判断某步骤是否可访问：第 0 步永远可访问；其他步骤需前
1675    一步已完成或自身已完成/跳过 */
1676export function isStepAccessible(step: WorkflowStepType, c
1677    ompletedSteps: WorkflowStepType[]): boolean {
1678  const idx = WORKFLOW_STEPS.findIndex(s => s.type === ste
1679    p);
1680  if (idx === 0) return true;
1681  const prev = WORKFLOW_STEPS[idx - 1].type;
1682  return completedSteps.includes(prev) || completedSteps.i
1683    ncludes(step);
1684}
1685
1686/** 获取步骤的元数据 */
1687export function getStepMeta(step: WorkflowStepType) {
1688  return WORKFLOW_STEPS.find(s => s.type === step);
1689}
1690
1691/** 获取步骤顺序索引 */
1692export function getStepOrder(step: WorkflowStepType): numb
1693    er {
1694  return WORKFLOW_STEPS.findIndex(s => s.type === step);
1695}
1696
1697// ========================================================
1698    =====
1699// 4. 步骤类型与 Prisma 枚举的映射
1700// ========================================================
```

```
  =====
export const stepTypeToPrisma: Record<WorkflowStepType, PrismaStepType> = {
  IDEATION: 'IDEATION',
  FRAMEWORK: 'FRAMEWORK',
  STYLE: 'STYLE',
  CHARACTER: 'CHARACTER',
  CONCEPT: 'CONCEPT',
  TRAILER: 'TRAILER',
  STORYBOARD: 'STORYBOARD',
  KEYFRAMES: 'KEYFRAMES',
  VIDEO_DIRECT: 'VIDEO_DIRECT',
  VIDEO_RENDER: 'VIDEO_RENDER',
  CAMERA: 'CAMERA',
  REVIEW: 'REVIEW',
};

/** 将 Prisma 枚举值转回应用层类型（用于从数据库读取后转换，包含历史隐藏步骤） */
export function prismaToStepType(prismaType: PrismaStepType): WorkflowStepType {
  return prismaType as WorkflowStepType;
}
/* ===== FILE: middleware.ts ===== */
import { NextResponse } from "next/server"
import type { NextRequest } from "next/server"
import { updateSession } from "@/lib/supabase/middleware"

// 受保护的路由前缀
const PROTECTED_PREFIXES = ["/dashboard", "/project", "/settings", "/admin"]

// 这些 API 路由也需要认证
const PROTECTED_API_PREFIXES = ["/api/projects", "/api/tasks"]

export async function middleware(request: NextRequest) {
  const { pathname } = request.nextUrl

  // 检查是否是受保护路由
  const isProtectedRoute = PROTECTED_PREFIXES.some((prefix) =>
    pathname.startsWith(prefix)
  )
  const isProtectedApi = PROTECTED_API_PREFIXES.some((prefix) =>
    pathname.startsWith(prefix)
  )

  // Supabase 模式：验证 session
```

```
try {
  const { response, user } = await updateSession(request
  )

  // 已登录且访问 /login → 重定向到 /dashboard
  if (user && pathname === "/login") {
    return NextResponse.redirect(new URL("/dashboard", r
  equest.url))
  }

  // 未登录且访问受保护路由 → 重定向到登录页
  if (!user && (isProtectedRoute || isProtectedApi)) {
    // API 路由返回 401
    if (isProtectedApi) {
      return NextResponse.json({ error: "AUTH_001" }, {
  status: 401 })
    }
    // 页面路由重定向到登录
    const loginUrl = new URL("/login", request.url)
    loginUrl.searchParams.set("redirect", pathname)
    return NextResponse.redirect(loginUrl)
  }

  return response
} catch (err) {
  // Supabase 配置异常（如环境变量缺失），未认证用户访问
  受保护路由时重定向
  console.error("[Middleware] Supabase session error:",
  err)
  if (!isProtectedRoute && !isProtectedApi) {
    return NextResponse.next()
  }

  // API 路由必须返回 JSON，不能返回 HTML 重定向，否则前
  端 res.json() 会解析失败
  if (isProtectedApi) {
    return NextResponse.json(
      { error: 'AUTH_001', message: '未登录或服务端认证
  配置异常' },
      { status: 401 }
    )
  }

  // 受保护页面路由在异常时保守处理：重定向到登录
  const loginUrl = new URL("/login", request.url)
  loginUrl.searchParams.set("redirect", pathname)
  return NextResponse.redirect(loginUrl)
}
}

```

```
export const config = {
  matcher: [
    "/dashboard",
    "/dashboard/:path*",
    "/project/:path*",
    "/settings/:path*",
    "/admin/:path*",
    "/api/projects/:path*",
    "/api/tasks/:path*",
    "/login",
  ],
}
/* ===== FILE: next.config.mjs ===== */
/** @type {import('next').NextConfig} */
const nextConfig = {
  env: {
    // Vercel 构建脚本会把静态 ffmpeg 下载到 public/ffmpeg
    // Next.js 运行时被部署到 /var/task，public 文件位于 /
    var/task/public/ffmpeg
    FFMPEG_PATH: './public/ffmpeg',
  },
  // 标记重型模块为外部依赖，不打包进 serverless function
  // 注意：lucide-react / recharts 不能同时出现在 transpil
    ePackages 和本列表中，故移除
  serverExternalPackages: [
    'bullmq',
    'sharp',
    '@img/sharp-libvips-linux-x64',
    'xlsx',
    'exceljs',
    '@aws-sdk/client-s3',
    '@aws-sdk/s3-request-presigner',
    'docx',
    'ffmpeg-static',
    'fluent-ffmpeg',
    'ioredis',
  ],
  images: {
    unoptimized: true,
    remotePatterns: [
      { protocol: 'https', hostname: '*.r2.cloudflarestora
    ge.com' },
      { protocol: 'https', hostname: '*.r2.dev' },
      { protocol: 'https', hostname: '*.supabase.co' },
    ],
  },
  // 现有代码库仍有大量历史 ESLint 债务：保持独立 lint 命
    令，避免与安全升级耦合。
  // TypeScript 构建检查保持启用。
  eslint: {
```

```
    ignoreDuringBuilds: true,
  },
  // 增加静态页面生成超时，避免单个页面卡住导致整体构建失败
  staticPageGenerationTimeout: 180,
}

export default nextConfig
/* ===== FILE: postcss.config.mjs ===== */
/** @type {import('postcss-load-config').Config} */
const config = {
  plugins: {
    tailwindcss: {},
    autoprefixer: {},
  },
};

export default config;
/* ===== FILE: prisma.config.ts ===== */
import { defineConfig } from 'prisma/config'

export default defineConfig({
  schema: './prisma/schema.prisma',
  migrations: {
    path: './prisma/migrations',
  },
  datasource: {
    // 迁移/数据库推送：必须使用直连 URL（如 Supabase 5432 端口）
    // 应用运行时通过 lib/prisma.ts 的 adapter 使用连接池 URL（如 Supabase 6543 端口）
    url: process.env.DIRECT_URL || process.env.DATABASE_URL || 'postgresql://placeholder@localhost/postgres',
  },
})
/* ===== FILE: prisma/migrations/20250612000000_add_video_segments/migration.sql ===== */
-- 工作指令.txt Phase 2：宣传片/直出视频分片重构所需的数据库变更
-- VideoSegment 表已在 schema.prisma 中定义，本迁移与 Prisma Schema 保持一致

-- 创建视频片段表
CREATE TABLE IF NOT EXISTS "VideoSegment" (
  id TEXT PRIMARY KEY,
  project_id TEXT NOT NULL REFERENCES "Project"(id) ON DELETE CASCADE,
  shot_id TEXT NOT NULL,
  step_name TEXT NOT NULL, -- 'TRAILER' | 'VIDEO_DIRECT' | 'VIDEO_RENDER'
```

```
  prompt TEXT NOT NULL,
  caption TEXT,
  video_url TEXT,
  storage_key TEXT,
  status TEXT NOT NULL DEFAULT 'pending', -- pending | gen
    erating | completed | failed
  sequence INTEGER NOT NULL,
  duration INTEGER,
  is_mock BOOLEAN NOT NULL DEFAULT false,
  error_message TEXT,
  created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAM
    P,
  updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAM
    P
);

CREATE INDEX IF NOT EXISTS "VideoSegment_project_id_status
    _idx" ON "VideoSegment"("project_id", "status");
CREATE INDEX IF NOT EXISTS "VideoSegment_project_id_step_n
    ame_idx" ON "VideoSegment"("project_id", "step_name");
CREATE INDEX IF NOT EXISTS "VideoSegment_project_id_sequen
    ce_idx" ON "VideoSegment"("project_id", "sequence");

-- 扩展项目表：合成视频与 BGM
ALTER TABLE "Project" ADD COLUMN IF NOT EXISTS combined_vi
    deo_url TEXT;
ALTER TABLE "Project" ADD COLUMN IF NOT EXISTS combined_vi
    deo_status TEXT DEFAULT 'pending';
ALTER TABLE "Project" ADD COLUMN IF NOT EXISTS bgm_url TEX
    T;
/* ===== FILE: prisma/migrations/20250717000000_add_voiceo
    ver_segments/migration.sql ===== */
-- CreateTable
CREATE TABLE "voiceover_segments" (
    "id" TEXT NOT NULL,
    "project_id" TEXT NOT NULL,
    "shot_id" TEXT NOT NULL,
    "step_name" TEXT NOT NULL,
    "text" TEXT NOT NULL,
    "speaker" TEXT,
    "audio_url" TEXT,
    "storage_key" TEXT,
    "status" TEXT NOT NULL DEFAULT 'pending',
    "sequence" INTEGER NOT NULL,
    "duration" INTEGER,
    "error_message" TEXT,
    "created_at" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIM
    ESTAMP,
    "updated_at" TIMESTAMP(3) NOT NULL,

```

```
    CONSTRAINT "voiceover_segments_pkey" PRIMARY KEY ("id"
    )
);

-- CreateIndex
CREATE INDEX "voiceover_segments_project_id_status_idx" ON
     "voiceover_segments"("project_id", "status");

-- CreateIndex
CREATE INDEX "voiceover_segments_project_id_step_name_idx"
     ON "voiceover_segments"("project_id", "step_name");

-- CreateIndex
CREATE INDEX "voiceover_segments_project_id_sequence_idx"
    ON "voiceover_segments"("project_id", "sequence");

-- CreateIndex
CREATE INDEX "voiceover_segments_project_id_shot_id_idx" O
    N "voiceover_segments"("project_id", "shot_id");

-- AddForeignKey
ALTER TABLE "voiceover_segments" ADD CONSTRAINT "voiceover
    _segments_project_id_fkey" FOREIGN KEY ("project_id")
    REFERENCES "Project"("id") ON DELETE CASCADE ON UPDATE
     CASCADE;
/* ===== FILE: prisma/migrations/20250717000001_add_voice_
    id_to_voiceover_segments/migration.sql ===== */
-- AlterTable
ALTER TABLE "voiceover_segments" ADD COLUMN "voice_id" TEX
    T;
/* ===== FILE: prisma/migrations/20260913000000_add_group_
    billing_audit/migration.sql ===== */
ALTER TABLE "OperationLog"
ADD COLUMN IF NOT EXISTS "billingSource" TEXT NOT NULL DEF
    AULT 'USER',
ADD COLUMN IF NOT EXISTS "billingGroupId" TEXT,
ADD COLUMN IF NOT EXISTS "balanceAfter" INTEGER;

CREATE INDEX IF NOT EXISTS "OperationLog_billingGroupId_cr
    eatedAt_idx"
ON "OperationLog"("billingGroupId", "createdAt");
/* ===== FILE: prisma/migrations/20260913000001_add_suppli
    er_operation_ledger/migration.sql ===== */
-- Additive supplier-operation ledger. Existing OperationL
    og rows remain readable.
ALTER TABLE "OperationLog"
  ADD COLUMN IF NOT EXISTS "actionKey" TEXT NOT NULL DEFAU
    LT 'generation.unknown',
  ADD COLUMN IF NOT EXISTS "category" TEXT NOT NULL DEFAUL
    T 'OTHER',
```

```
ADD COLUMN IF NOT EXISTS "status" TEXT NOT NULL DEFAULT
  'SUBMITTED',
ADD COLUMN IF NOT EXISTS "stepName" TEXT,
ADD COLUMN IF NOT EXISTS "providerCost" DECIMAL(18,8),
ADD COLUMN IF NOT EXISTS "currency" TEXT,
ADD COLUMN IF NOT EXISTS "costSource" TEXT NOT NULL DEFA
  ULT 'UNKNOWN',
ADD COLUMN IF NOT EXISTS "idempotencyKey" TEXT,
ADD COLUMN IF NOT EXISTS "inputSummary" JSONB,
ADD COLUMN IF NOT EXISTS "metadata" JSONB,
ADD COLUMN IF NOT EXISTS "errorCode" TEXT,
ADD COLUMN IF NOT EXISTS "startedAt" TIMESTAMP(3) NOT NU
  LL DEFAULT CURRENT_TIMESTAMP,
ADD COLUMN IF NOT EXISTS "completedAt" TIMESTAMP(3),
ADD COLUMN IF NOT EXISTS "durationMs" INTEGER,
ADD COLUMN IF NOT EXISTS "updatedAt" TIMESTAMP(3) NOT NU
  LL DEFAULT CURRENT_TIMESTAMP;

UPDATE "OperationLog"
SET "status" = CASE WHEN "success" THEN 'SUCCEEDED' ELSE '
  FAILED' END,
  "completedAt" = COALESCE("completedAt", "createdAt"),
  "startedAt" = COALESCE("startedAt", "createdAt"),
  "actionKey" = CASE WHEN "type" = 'regenerate' THEN 'ge
  neration.regenerate' WHEN "type" = 'error' THEN 'gener
  ation.failed' ELSE 'generation.generate' END
WHERE "actionKey" = 'generation.unknown';

CREATE TABLE IF NOT EXISTS "ProviderCallAttempt" (
 "id" TEXT NOT NULL, "operationId" TEXT NOT NULL, "provid
  er" TEXT NOT NULL,
 "model" TEXT, "endpoint" TEXT NOT NULL, "method" TEXT NO
  T NULL DEFAULT 'POST',
 "attemptNo" INTEGER NOT NULL DEFAULT 1, "status" TEXT NO
  T NULL DEFAULT 'RUNNING',
 "externalTaskId" TEXT, "requestId" TEXT, "httpStatus" IN
  TEGER,
 "pollCount" INTEGER NOT NULL DEFAULT 0, "providerCost" D
  ECIMAL(18,8),
 "currency" TEXT, "costSource" TEXT NOT NULL DEFAULT 'UNK
  NOWN',
 "errorCode" TEXT, "errorMessage" TEXT, "metadata" JSONB,
 "startedAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMEST
  AMP,
 "completedAt" TIMESTAMP(3), "durationMs" INTEGER,
 CONSTRAINT "ProviderCallAttempt_pkey" PRIMARY KEY ("id")
  ,
 CONSTRAINT "ProviderCallAttempt_operationId_fkey" FOREIG
  N KEY ("operationId") REFERENCES "OperationLog"("id")
  ON DELETE CASCADE ON UPDATE CASCADE
```

```
);

CREATE TABLE IF NOT EXISTS "OperationResult" (
  "id" TEXT NOT NULL, "operationId" TEXT NOT NULL, "kind"
    TEXT NOT NULL,
  "title" TEXT, "assetId" TEXT, "userAssetId" TEXT, "video
    SegmentId" TEXT,
  "voiceoverSegmentId" TEXT, "storageKey" TEXT, "mimeType"
     TEXT,
  "externalUrl" TEXT, "isMock" BOOLEAN NOT NULL DEFAULT fa
    lse, "metadata" JSONB,
  "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMEST
    AMP,
  CONSTRAINT "OperationResult_pkey" PRIMARY KEY ("id"),
  CONSTRAINT "OperationResult_operationId_fkey" FOREIGN KE
    Y ("operationId") REFERENCES "OperationLog"("id") ON D
    ELETE CASCADE ON UPDATE CASCADE
);

CREATE UNIQUE INDEX IF NOT EXISTS "OperationLog_idempotenc
    yKey_key" ON "OperationLog"("idempotencyKey");
CREATE INDEX IF NOT EXISTS "OperationLog_status_createdAt_
    idx" ON "OperationLog"("status", "createdAt");
CREATE INDEX IF NOT EXISTS "OperationLog_projectId_created
    At_idx" ON "OperationLog"("projectId", "createdAt");
CREATE INDEX IF NOT EXISTS "OperationLog_category_createdA
    t_idx" ON "OperationLog"("category", "createdAt");
CREATE INDEX IF NOT EXISTS "ProviderCallAttempt_operationI
    d_startedAt_idx" ON "ProviderCallAttempt"("operationId
    ", "startedAt");
CREATE INDEX IF NOT EXISTS "ProviderCallAttempt_provider_m
    odel_startedAt_idx" ON "ProviderCallAttempt"("provider
    ", "model", "startedAt");
CREATE INDEX IF NOT EXISTS "ProviderCallAttempt_externalTa
    skId_idx" ON "ProviderCallAttempt"("externalTaskId");
CREATE INDEX IF NOT EXISTS "OperationResult_operationId_cr
    eatedAt_idx" ON "OperationResult"("operationId", "crea
    tedAt");
CREATE UNIQUE INDEX IF NOT EXISTS "OperationResult_operati
    onId_assetId_key" ON "OperationResult"("operationId",
    "assetId");
CREATE UNIQUE INDEX IF NOT EXISTS "OperationResult_operati
    onId_userAssetId_key" ON "OperationResult"("operationI
    d", "userAssetId");
CREATE UNIQUE INDEX IF NOT EXISTS "OperationResult_operati
    onId_videoSegmentId_key" ON "OperationResult"("operati
    onId", "videoSegmentId");
CREATE UNIQUE INDEX IF NOT EXISTS "OperationResult_operati
    onId_voiceoverSegmentId_key" ON "OperationResult"("ope
    rationId", "voiceoverSegmentId");
```

```
/* ===== FILE: prisma/migrations/20260913000002_add_supplier_usage_reconciliation/migration.sql ===== */
-- Preserve measurable usage independently from money. OpenLux-compatible
-- generation responses expose token usage, but do not consistently expose the
-- account's final charged amount. Actual cost is reconciled separately.
ALTER TABLE "ProviderCallAttempt"
  ADD COLUMN IF NOT EXISTS "inputTokens" INTEGER,
  ADD COLUMN IF NOT EXISTS "outputTokens" INTEGER,
  ADD COLUMN IF NOT EXISTS "totalTokens" INTEGER,
  ADD COLUMN IF NOT EXISTS "reconciledAt" TIMESTAMP(3);

ALTER TABLE "OperationLog"
  ADD COLUMN IF NOT EXISTS "pointsRefunded" INTEGER NOT NULL DEFAULT 0;

CREATE INDEX IF NOT EXISTS "ProviderCallAttempt_requestId_idx"
  ON "ProviderCallAttempt"("requestId");
/* ===== FILE: prisma/migrations/20260914000000_add_group_contribution_tracking/migration.sql ===== */
-- Normalize the business target and adoption state used by group contribution reports.
ALTER TABLE "OperationLog"
  ADD COLUMN IF NOT EXISTS "scopeType" TEXT,
  ADD COLUMN IF NOT EXISTS "scopeKey" TEXT;

ALTER TABLE "OperationResult"
  ADD COLUMN IF NOT EXISTS "targetType" TEXT,
  ADD COLUMN IF NOT EXISTS "targetKey" TEXT,
  ADD COLUMN IF NOT EXISTS "targetLabel" TEXT,
  ADD COLUMN IF NOT EXISTS "shotId" TEXT,
  ADD COLUMN IF NOT EXISTS "actNumber" INTEGER,
  ADD COLUMN IF NOT EXISTS "adoptionStatus" TEXT NOT NULL DEFAULT 'GENERATED',
  ADD COLUMN IF NOT EXISTS "adoptedAt" TIMESTAMP(3),
  ADD COLUMN IF NOT EXISTS "adoptedById" TEXT;

CREATE INDEX IF NOT EXISTS "OperationLog_projectId_scopeType_scopeKey_idx"
  ON "OperationLog"("projectId", "scopeType", "scopeKey");
CREATE INDEX IF NOT EXISTS "OperationResult_targetType_targetKey_idx"
  ON "OperationResult"("targetType", "targetKey");
CREATE INDEX IF NOT EXISTS "OperationResult_adoptionStatus_createdAt_idx"
  ON "OperationResult"("adoptionStatus", "createdAt");

```

```
-- Backfill normalized shot dimensions from the immutable result metadata snapshot.
UPDATE "OperationResult"
SET
  "targetType" = COALESCE("targetType", 'SHOT'),
  "shotId" = COALESCE("shotId", "metadata"->>'shotId'),
  "actNumber" = COALESCE(
    "actNumber",
    CASE
      WHEN ("metadata"->>'actNumber') ~ '^[0-9]+$'
      THEN ("metadata"->>'actNumber')::INTEGER
      ELSE NULL
    END
  ),
  "targetKey" = COALESCE(
    "targetKey",
    CASE
      WHEN "metadata"->>'shotId' IS NOT NULL
      THEN 'act:' || COALESCE("metadata"->>'actNumber', '0') || '/shot:' || ("metadata"->>'shotId')
      ELSE NULL
    END
  )
WHERE "metadata"->>'shotId' IS NOT NULL;
/* ===== FILE: prisma/migrations/20260914000001_add_asset_contribution_attribution/migration.sql ===== */
ALTER TABLE "Asset"
  ADD COLUMN IF NOT EXISTS "createdById" TEXT,
  ADD COLUMN IF NOT EXISTS "origin" TEXT NOT NULL DEFAULT 'GENERATED';

ALTER TABLE "VideoSegment"
  ADD COLUMN IF NOT EXISTS "created_by_id" TEXT;

CREATE INDEX IF NOT EXISTS "Asset_createdById_createdAt_idx"
  ON "Asset"("createdById", "createdAt");
CREATE INDEX IF NOT EXISTS "VideoSegment_createdById_createdAt_idx"
  ON "VideoSegment"("created_by_id", "created_at");

-- Prefer the actor captured by the immutable operation ledger.
UPDATE "Asset" AS asset
SET "createdById" = operation."userId"
FROM "OperationResult" AS result
JOIN "OperationLog" AS operation ON operation."id" = result."operationId"
WHERE result."assetId" = asset."id"
  AND asset."createdById" IS NULL;
```

```
2201
2202-- Older generation routes stored the result directly on O
2203    perationLog.assetId.
2204UPDATE "Asset" AS asset
2205SET "createdById" = operation."userId"
2206FROM "OperationLog" AS operation
2207WHERE operation."assetId" = asset."id"
2208  AND asset."createdById" IS NULL;
2209
2210-- Imported storyboard images did not historically have an
2211     operation row. Their
2212-- best recoverable owner is the project creator; new impo
2213    rts record the actor exactly.
2214UPDATE "Asset" AS asset
2215SET
2216  "createdById" = project."userId"
2217FROM "Project" AS project
2218WHERE project."id" = asset."projectId"
2219  AND asset."createdById" IS NULL;
2220
2221UPDATE "Asset"
2222SET "origin" = 'IMPORTED'
2223WHERE "metadata"->>'source' = 'storyboard-import';
2224
2225UPDATE "VideoSegment" AS segment
2226SET "created_by_id" = COALESCE(
2227  (
2228    SELECT operation."userId"
2229    FROM "OperationResult" AS result
2230    JOIN "OperationLog" AS operation ON operation."id" = r
2231    esult."operationId"
2232    WHERE result."videoSegmentId" = segment."id"
2233    ORDER BY result."createdAt" DESC
2234    LIMIT 1
2235  ),
2236  project."userId"
2237)
2238FROM "Project" AS project
2239WHERE project."id" = segment."project_id"
2240  AND segment."created_by_id" IS NULL;
2241/* ===== FILE: prisma/migrations/20260915000000_add_workfl
2242    owstep_updated_at/migration.sql ===== */
2243-- WorkflowStep.updatedAt：并发局部更新 outputData 所需的
2244    乐观锁 CAS 字段。
2245--
2246-- 背景：outputData 是 JSON 列，生成流程普遍是「读快照 →
2247    跑 30~240s 的生成 →
2248-- 整块写回」。并发生成不同镜头时，后写的那次会用旧快照覆
2249    盖前一次的结果，
2250-- 对方的 firstFrameUrl / shotAssets / generatingShots 条
```

```
  目凭空消失，
-- 卡片永远卡在「生成中」。lib/workflow-executor.mergeStep
  OutputData 用
-- `UPDATE ... WHERE id = ? AND updated_at = ?` 做 CAS，冲
  突则重读重算。
--
-- ⚠️ 必须先应用本迁移再上线：mergeStepOutputData / tryM
  ergeStepOutputData
-- 会 select 和 where updated_at，列不存在时相关接口会直接
  报 500。
-- 应用方式：npx prisma migrate deploy
ALTER TABLE "WorkflowStep"
  ADD COLUMN IF NOT EXISTS "updated_at" TIMESTAMP(3) NOT N
  ULL DEFAULT CURRENT_TIMESTAMP;

CREATE INDEX IF NOT EXISTS "WorkflowStep_projectId_updated
  At_idx"
  ON "WorkflowStep"("projectId", "updated_at");
/* ===== FILE: prisma/schema.prisma ===== */
generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider = "postgresql"
}

// 项目状态
enum ProjectStatus {
  ACTIVE
  ARCHIVED
  DELETED
}

enum GroupRole {
  ADMIN
  MEMBER
}

enum MembershipStatus {
  ACTIVE
  INVITED
  PENDING
}

enum GroupCostMode {
  MEMBER_PAY
  GROUP_POOL
}

```

```
// 12 步工作流类型
enum WorkflowStepType {
  IDEATION // 创意扩散
  FRAMEWORK // 框架搭建
  STYLE // 风格统一
  CHARACTER // 人物设计
  CONCEPT // 概念图生成
  TRAILER // 30s 宣传片
  STORYBOARD // 分镜设计
  KEYFRAMES // 首尾帧
  VIDEO_DIRECT // 直生视频
  VIDEO_RENDER // AI 渲染
  CAMERA // 电脑运镜
  REVIEW // 一致性检测/评测
}

// 步骤执行状态
enum StepStatus {
  PENDING
  PROCESSING
  COMPLETED
  FAILED
  SKIPPED
}

// 资源类型
enum AssetType {
  IMAGE
  VIDEO
  TEXT
  AUDIO
  REFERENCE
}

enum UserAssetKind {
  CHARACTER
  ENVIRONMENT
  REFERENCE
}

enum UserAssetSource {
  UPLOAD
  GENERATED
}

// 评测维度
enum EvaluationDimension {
  CONSISTENCY // 时序一致性
  FIDELITY // 语义保真度
  AESTHETICS // 视觉美学
```

```
  SPEED // 生成时效
}

// NextAuth.js v5 所需的 Account 模型
model Account {
  id                String  @id @default(cuid())
  userId            String
  type              String
  provider          String
  providerAccountId String
  refresh_token     String? @db.Text
  access_token      String? @db.Text
  expires_at        Int?
  token_type        String?
  scope             String?
  id_token          String? @db.Text
  session_state     String?
  user              User    @relation(fields: [userId], re
    ferences: [id], onDelete: Cascade)

  @@unique([provider, providerAccountId])
}

// NextAuth.js v5 所需的 Session 模型
model Session {
  id           String   @id @default(cuid())
  sessionToken String   @unique
  userId       String
  expires      DateTime
  user         User     @relation(fields: [userId], refere
    nces: [id], onDelete: Cascade)
}

// NextAuth.js v5 所需的 VerificationToken 模型（Email Mag
    ic Link 用）
model VerificationToken {
  identifier String
  token      String   @unique
  expires    DateTime

  @@unique([identifier, token])
}

model User {
  id                String               @id @default(cui
    d())
  email             String               @unique
  name              String?
  image             String?
  emailVerified     DateTime?
```

```
  points             Int                 @default(0)
  isAdmin            Boolean             @default(false)
  adminNote          String?
  accounts           Account[]
  sessions           Session[]
  projects           Project[]
  userAssets         UserAsset[]
  auditLogs          AuditLog[]
  operationLogs      OperationLog[]
  rechargeOrders     RechargeOrder[]
  creativeIterations CreativeIteration[]
  groupMemberships   GroupMembership[]
  createdGroups      Group[]              @relation("Creat
    edGroups")
  invitedMemberships GroupMembership[]    @relation("Invit
    edBy")
  pointTransfersOut  GroupPointTransfer[] @relation("Trans
    ferFromUser")
  createdAt          DateTime             @default(now())
}

model UserAsset {
  id          String          @id @default(cuid())
  userId      String          @map("user_id")
  user        User            @relation(fields: [userId],
    references: [id], onDelete: Cascade)
  kind        UserAssetKind
  source      UserAssetSource
  title       String?
  description String?         @db.Text
  prompt      String?         @db.Text
  templateId  String?         @map("template_id")
  mimeType    String          @map("mime_type")
  storageKey  String          @map("storage_key")
  url         String
  metadata    Json?
  createdAt   DateTime        @default(now()) @map("create
    d_at")
  updatedAt   DateTime        @updatedAt @map("updated_at"
    )

  @@index([userId, kind, createdAt])
  @@map("user_assets")
}

model Project {
  id                String        @id @default(cuid())
  userId            String
  user              User          @relation(fields: [userI
    d], references: [id], onDelete: Cascade)
```

```
  title             String        @default("未命名项目")
  rawIdea           String        @db.Text
  selectedDirection Json? // 选定的创意扩散方向
  framework         Json? // 框架搭建结果（背景/风格/角色/
    梗概/幕结构）
  selectedStyleId   String? // 关联 Asset.id
  status            ProjectStatus @default(ACTIVE)

  // 框架导入相关字段
  frameworkSource  String? @default("generated") @map("fra
    mework_source") // 'generated' | 'imported' | 'mixed'
  importedFileUrl  String? @map("imported_file_url")
  importedFileName String? @map("imported_file_name")
  importedRawText  String? @map("imported_raw_text") @db.T
    ext

  // 小组关联
  groupId String? @map("group_id")
  group   Group?  @relation(fields: [groupId], references:
     [id], onDelete: SetNull)

  // 9 步工作流完成状态（条件解锁 DAG 状态机）
  stepIdeaDone                Boolean @default(false) @ma
    p("step_idea_done")
  stepFrameworkDone           Boolean @default(false) @ma
    p("step_framework_done")
  stepStyleDone               Boolean @default(false) @ma
    p("step_style_done")
  stepCharacterDone           Boolean @default(false) @ma
    p("step_character_done")
  stepConceptDone             Boolean @default(false) @ma
    p("step_concept_done")
  stepStoryboardDone          Boolean @default(false) @ma
    p("step_storyboard_done")
  stepStoryboardFirstframeDone Boolean @default(false) @ma
    p("step_storyboard_firstframe_done")
  stepTrailerDone             Boolean @default(false) @ma
    p("step_trailer_done")
  stepEndingDone              Boolean @default(false) @ma
    p("step_ending_done") // KEYFRAMES
  stepDirectDone              Boolean @default(false) @ma
    p("step_direct_done") // VIDEO_DIRECT

  steps              WorkflowStep[]
  assets             Asset[]
  comparisons        Comparison[]
  creativeIterations CreativeIteration[]
  videoSegments      VideoSegment[]
  voiceoverSegments  VoiceoverSegment[]

```

```
  createdAt DateTime @default(now())
  updatedAt DateTime @updatedAt

  @@index([userId, status])
  @@index([groupId])
}

// 充值订单
model RechargeOrder {
  id            String   @id @default(cuid())
  userId        String
  user          User     @relation(fields: [userId], refer
    ences: [id], onDelete: Cascade)
  amountYuan    Int      @map("amount_yuan") // 人民币金额
    （分）
  points        Int // 对应点数
  paymentMethod String   @default("bank_transfer") @map("p
    ayment_method") // bank_transfer | alipay | wechat
  proofImageUrl String?  @map("proof_image_url") // 转账凭
    证图片 URL
  status        String   @default("pending") // pending |
    approved | rejected
  adminNote     String?  @map("admin_note") // 管理员审核
    备注
  createdAt     DateTime @default(now()) @map("created_at"
    )
  updatedAt     DateTime @updatedAt @map("updated_at")

  @@index([userId, status])
  @@index([status, createdAt])
  @@map("recharge_orders")
}

model Group {
  id             String               @id @default(cuid())
  name           String
  description    String?              @db.Text
  createdById    String               @map("created_by_id"
    )
  createdBy      User                 @relation("CreatedGr
    oups", fields: [createdById], references: [id], onDele
    te: Cascade)
  points         Int                  @default(0)
  costMode       GroupCostMode        @default(MEMBER_PAY)
     @map("cost_mode")
  members        GroupMembership[]
  projects       Project[]
  pointTransfers GroupPointTransfer[]
  createdAt      DateTime             @default(now())
  updatedAt      DateTime             @updatedAt
```

```
2551
2552  @@index([createdById])
2553}
2554
2555model GroupMembership {
2556  id          String           @id @default(cuid())
2557  groupId     String           @map("group_id")
2558  group       Group            @relation(fields: [groupId]
2559    , references: [id], onDelete: Cascade)
2560  userId      String           @map("user_id")
2561  user        User             @relation(fields: [userId],
2562     references: [id], onDelete: Cascade)
2563  role        GroupRole        @default(MEMBER)
2564  status      MembershipStatus @default(ACTIVE)
2565  invitedById String?          @map("invited_by_id")
2566  invitedBy   User?            @relation("InvitedBy", fiel
2567    ds: [invitedById], references: [id], onDelete: SetNull
2568    )
2569  joinedAt    DateTime         @default(now()) @map("joine
2570    d_at")
2571  createdAt   DateTime         @default(now())
2572
2573  @@unique([groupId, userId])
2574  @@index([userId, status])
2575  @@map("group_memberships")
2576}
2577
2578model GroupPointTransfer {
2579  id         String   @id @default(cuid())
2580  groupId    String   @map("group_id")
2581  group      Group    @relation(fields: [groupId], referen
2582    ces: [id], onDelete: Cascade)
2583  userId     String   @map("user_id")
2584  user       User     @relation("TransferFromUser", fields
2585    : [userId], references: [id], onDelete: Cascade)
2586  amount     Int
2587  userAfter  Int      @map("user_after")
2588  groupAfter Int      @map("group_after")
2589  createdAt  DateTime @default(now())
2590
2591  @@index([groupId])
2592  @@index([userId])
2593  @@map("group_point_transfers")
2594}
2595
2596model WorkflowStep {
2597  id           String           @id @default(cuid())
2598  projectId    String
2599  project      Project          @relation(fields: [project
2600    Id], references: [id], onDelete: Cascade)
```

```
  stepType     WorkflowStepType
  status       StepStatus       @default(PENDING)
  inputData    Json? // 该步输入参数
  outputData   Json? // 该步输出结果
  errorMessage String?          @db.Text
  order        Int // 步骤顺序（0-11）
  retryCount   Int              @default(0)
  startedAt    DateTime?
  completedAt  DateTime?
  resultAssets Asset[] // 关联生成的资源
  // 乐观锁 CAS 字段：并发局部更新 outputData 时用于检测期
    间是否有别的写入
  // 见 lib/workflow-executor.mergeStepOutputData
  updatedAt    DateTime         @updatedAt @map("updated_a
    t")
  createdAt    DateTime         @default(now())

  @@unique([projectId, stepType]) // 一个项目同一步骤只能
    有一条记录
  @@index([projectId, order])
}

model Asset {
  id          String        @id @default(cuid())
  projectId   String
  project     Project       @relation(fields: [projectId],
     references: [id], onDelete: Cascade)
  stepId      String?
  step        WorkflowStep? @relation(fields: [stepId], re
    ferences: [id], onDelete: SetNull)
  type        AssetType
  mimeType    String
  storageKey  String // R2 上的唯一 key
  url         String // 预签名 URL（短期）
  metadata    Json? // { seed, modelUsed, prompt, duration
    , size, styleRef, characterRef }
  createdById String?
  origin      String        @default("GENERATED")
  createdAt   DateTime      @default(now())

  @@index([projectId, type])
  @@index([stepId])
  @@index([createdById, createdAt])
}

model Evaluation {
  id           String              @id @default(cuid())
  assetId      String
  comparisonId String?
  dimension    EvaluationDimension
```

```
  aiScore      Float? // 自动评测 1-10
  humanScore   Float? // 人工评分 1-10
  details      Json? // 帧级标记、差异说明等
  createdAt    DateTime            @default(now())

  @@index([assetId, dimension])
}

model Comparison {
  id             String   @id @default(cuid())
  projectId      String
  project        Project  @relation(fields: [projectId], r
    eferences: [id], onDelete: Cascade)
  scriptSnapshot Json // 评测时的分镜脚本快照
  modelResults   Json // { modelName: { taskId, assetId, s
    tatus, url, cost } }
  overallScores  Json? // 各模型总分
  status         String   @default("PENDING") // PENDING,
    PROCESSING, COMPLETED
  createdAt      DateTime @default(now())

  @@index([projectId, status])
}

model AuditLog {
  id        String   @id @default(cuid())
  userId    String?
  user      User?    @relation(fields: [userId], reference
    s: [id], onDelete: SetNull)
  action    String // GENERATE_IMAGE, GENERATE_VIDEO, LOGI
    N, EXPORT 等
  inputHash String? // 输入提示词的 SHA256
  outputUrl String?
  modelUsed String?
  ipAddress String?
  userAgent String?
  errorCode String?
  createdAt DateTime @default(now())

  @@index([userId, createdAt])
  @@index([action, createdAt])
}

model OperationLog {
  id               String               @id @default(uuid
    ())
  userId           String
  user             User                 @relation(fields:
     [userId], references: [id], onDelete: Cascade)
  type             String // legacy: generate, regenerate,
```

```
    error
  actionKey        String               @default("generat
    ion.unknown")
  category         String               @default("OTHER")
  status           String               @default("SUBMITT
    ED")
  projectId        String?
  workflowStepId   String?
  stepName         String?
  scopeType        String?
  scopeKey         String?
  assetId          String? // legacy polymorphic result id
  pointsCost       Int                   @default(0)
  pointsRefunded   Int                   @default(0)
  success          Boolean               @default(true)
  errorCode        String?
  errorMessage     String?
  billingSource    String                @default("USER")
  billingGroupId   String?
  balanceAfter     Int?
  providerCost     Decimal?              @db.Decimal(18, 8
    )
  currency         String?
  costSource       String                @default("UNKNOWN
    ")
  idempotencyKey   String?               @unique
  inputSummary     Json?
  metadata         Json?
  startedAt        DateTime              @default(now())
  completedAt      DateTime?
  durationMs       Int?
  createdAt        DateTime              @default(now())
  updatedAt        DateTime              @updatedAt
  providerAttempts ProviderCallAttempt[]
  results          OperationResult[]

  @@index([userId, createdAt])
  @@index([billingGroupId, createdAt])
  @@index([type, createdAt])
  @@index([status, createdAt])
  @@index([projectId, createdAt])
  @@index([projectId, scopeType, scopeKey])
  @@index([category, createdAt])
}

model ProviderCallAttempt {
  id             String       @id @default(uuid())
  operationId    String
  operation      OperationLog @relation(fields: [operation
    Id], references: [id], onDelete: Cascade)
```

```
  provider       String
  model          String?
  endpoint       String
  method         String       @default("POST")
  attemptNo      Int          @default(1)
  status         String       @default("RUNNING")
  externalTaskId String?
  requestId      String?
  httpStatus     Int?
  pollCount      Int          @default(0)
  providerCost   Decimal?     @db.Decimal(18, 8)
  currency       String?
  costSource     String       @default("UNKNOWN")
  inputTokens    Int?
  outputTokens   Int?
  totalTokens    Int?
  reconciledAt   DateTime?
  errorCode      String?
  errorMessage   String?
  metadata       Json?
  startedAt      DateTime     @default(now())
  completedAt    DateTime?
  durationMs     Int?

  @@index([operationId, startedAt])
  @@index([provider, model, startedAt])
  @@index([externalTaskId])
  @@index([requestId])
}

model OperationResult {
  id                String       @id @default(uuid())
  operationId       String
  operation         OperationLog @relation(fields: [opera
    tionId], references: [id], onDelete: Cascade)
  kind              String
  title             String?
  assetId           String?
  userAssetId       String?
  videoSegmentId    String?
  voiceoverSegmentId String?
  storageKey        String?
  mimeType          String?
  externalUrl       String?
  isMock            Boolean      @default(false)
  targetType        String?
  targetKey         String?
  targetLabel       String?
  shotId            String?
  actNumber         Int?
```

```
  adoptionStatus     String       @default("GENERATED")
  adoptedAt          DateTime?
  adoptedById        String?
  metadata           Json?
  createdAt          DateTime     @default(now())

  @@unique([operationId, assetId])
  @@unique([operationId, userAssetId])
  @@unique([operationId, videoSegmentId])
  @@unique([operationId, voiceoverSegmentId])
  @@index([operationId, createdAt])
  @@index([targetType, targetKey])
  @@index([adoptionStatus, createdAt])
}

// 创意扩散深化迭代历史版本
model VideoSegment {
  id           String   @id @default(cuid())
  projectId    String   @map("project_id")
  project      Project  @relation(fields: [projectId], ref
    erences: [id], onDelete: Cascade)
  shotId       String   @map("shot_id") // 对应分镜 shot_0
    01 / 概念图 asset id
  actNumber    Int?     @map("act_number") // VIDEO_DIRECT
     中区分不同幕的同名 shotId
  stepName     String   @map("step_name") // 'TRAILER' | '
    VIDEO_DIRECT' | 'VIDEO_RENDER'
  prompt       String // 英文视频生成提示词
  caption      String?  @map("caption") // 中文画面描述（
    前端展示用）
  videoUrl     String?  @map("video_url") // 生成后的视频
    URL（R2 签名 URL）
  storageKey   String?  @map("storage_key") // R2 存储 key
  status       String   @default("pending") @map("status")
     // pending | generating | completed | failed
  sequence     Int      @map("sequence") // 片段顺序（用于
     ffmpeg concat）
  duration     Int?     @map("duration") // 片段时长（秒）
  isMock       Boolean  @default(false) @map("is_mock") //
     是否 Ken Burns 兜底
  errorMessage String?  @map("error_message") // 失败原因
  createdById  String?  @map("created_by_id")
  createdAt    DateTime @default(now()) @map("created_at")
  updatedAt    DateTime @updatedAt @map("updated_at")

  @@index([projectId, status])
  @@index([projectId, stepName])
  @@index([projectId, sequence])
  @@index([projectId, stepName, actNumber, shotId])
  @@index([createdById, createdAt])
```

```
2851  @@map("VideoSegment")
2852}
2853
2854model CreativeIteration {
2855  id                  String   @id @default(cuid())
2856  projectId           String   @map("project_id")
2857  project             Project  @relation(fields: [projectI
2858    d], references: [id], onDelete: Cascade)
2859  userId              String   @map("user_id")
2860  user                User     @relation(fields: [userId],
2861     references: [id], onDelete: Cascade)
2862  versionNumber       Int      @map("version_number")
2863  originalInput       String   @map("original_input") @db.
2864    Text
2865  creativeContent     String   @map("creative_content") @d
2866    b.Text
2867  retentionScore      Int?     @map("retention_score") //
2868    原内容保留度 0-100
2869  qualityScore        Int?     @map("quality_score") // 优
2870    秀程度 0-100
2871  concerns            String?  @db.Text // AI 指出的核心顾
2872    虑
2873  improvementOptions  Json?    @map("improvement_options")
2874     // 改进选项数组
2875  selectedImprovement String?  @map("selected_improvement"
2876    ) // 用户选中的选项
2877  customFeedback      String?  @map("custom_feedback") @db
2878    .Text
2879  isCurrent           Boolean  @default(false) @map("is_cu
2880    rrent")
2881  createdAt           DateTime @default(now()) @map("creat
2882    ed_at")
2883
2884  // 注意：isCurrent 的唯一性由应用层保证（updateMany + cr
2885    eate 的原子操作）
2886  @@index([projectId, versionNumber])
2887  @@map("creative_iterations")
2888}
2889
2890model VoiceoverSegment {
2891  id           String   @id @default(cuid())
2892  projectId    String   @map("project_id")
2893  project      Project  @relation(fields: [projectId], ref
2894    erences: [id], onDelete: Cascade)
2895  shotId       String   @map("shot_id") // 对应分镜 shotId
2896  stepName     String   @map("step_name") // 'VIDEO_DIRECT
2897    ' | 'TRAILER'
2898  text         String   @db.Text // 配音文案/旁白
2899  speaker      String? // 说话人/角色名（如"旁白"）
2900  voiceId      String?  @map("voice_id") // MiniMax 音色 I
```

```
  D（AI 按角色人设自动分配）
audioUrl     String?  @map("audio_url") // 生成后的音频
  URL（R2 签名 URL）
storageKey   String?  @map("storage_key") // R2 存储 key
status       String   @default("pending") @map("status")
   // pending | generating | completed | failed
sequence     Int      @map("sequence") // 配音顺序
duration     Int?     @map("duration") // 音频时长（毫秒
  ）
errorMessage String?  @map("error_message") // 失败原因
createdAt    DateTime @default(now()) @map("created_at")
updatedAt    DateTime @updatedAt @map("updated_at")

@@index([projectId, status])
@@index([projectId, stepName])
@@index([projectId, sequence])
@@index([projectId, shotId])
@@map("voiceover_segments")
}
/* ===== FILE: tailwind.config.ts ===== */
import type { Config } from "tailwindcss"

const config: Config = {
 darkMode: ["class"],
 content: [
   "./pages/**/*.{js,ts,jsx,tsx,mdx}",
   "./components/**/*.{js,ts,jsx,tsx,mdx}",
   "./app/**/*.{js,ts,jsx,tsx,mdx}",
 ],
 theme: {
   extend: {
     colors: {
       border: "hsl(var(--border))",
       input: "hsl(var(--input))",
       ring: "hsl(var(--ring))",
       background: "hsl(var(--background))",
       foreground: "hsl(var(--foreground))",
       primary: {
         DEFAULT: "hsl(var(--primary))",
         foreground: "hsl(var(--primary-foreground))",
       },
       secondary: {
         DEFAULT: "hsl(var(--secondary))",
         foreground: "hsl(var(--secondary-foreground))",
       },
       destructive: {
         DEFAULT: "hsl(var(--destructive))",
         foreground: "hsl(var(--destructive-foreground))"
   ,
       },
```

```
      muted: {
        DEFAULT: "hsl(var(--muted))",
        foreground: "hsl(var(--muted-foreground))",
      },
      accent: {
        DEFAULT: "hsl(var(--accent))",
        foreground: "hsl(var(--accent-foreground))",
      },
      popover: {
        DEFAULT: "hsl(var(--popover))",
        foreground: "hsl(var(--popover-foreground))",
      },
      card: {
        DEFAULT: "hsl(var(--card))",
        foreground: "hsl(var(--card-foreground))",
      },
    },
    borderRadius: {
      lg: "var(--radius)",
      md: "calc(var(--radius) - 2px)",
      sm: "calc(var(--radius) - 4px)",
    },
  },
},
plugins: [require("tailwindcss-animate")],
}
export default config
/* ===== FILE: types/index.ts ===== */
export type AssetType = 'IMAGE' | 'VIDEO' | 'TEXT' | 'AUDI
   O' | 'REFERENCE';

export type ProjectStatus = 'ACTIVE' | 'ARCHIVED' | 'DELET
   ED';

export type WorkflowStepType =
| 'IDEATION'
| 'FRAMEWORK'
| 'STYLE'
| 'CHARACTER'
| 'CONCEPT'
| 'TRAILER'
| 'STORYBOARD'
| 'KEYFRAMES'
| 'VIDEO_DIRECT'
| 'VIDEO_RENDER'
| 'CAMERA'
| 'REVIEW';

export type StepStatus = 'PENDING' | 'PROCESSING' | 'COMPL
   ETED' | 'FAILED' | 'SKIPPED';
```